AnD AI (Task B): A Stateless Agentic Framework for Culturally-Aware Recommendations

James Ayomide Adewara, Esther Omole, Fedora Anaba, Ifeoluwa Sotola

Abstract

Traditional recommendation systems treat users as static profiles, failing in emerging markets where decisions are driven by inflation, location logistics, and local language. We introduce AnD AI (Task B), a stateless, persona-driven recommendation agent that simulates the Nigerian consumer as an economic agent. Evaluated on a hybrid catalog of 173 Nigerian-localized items, Task B achieves NDCG@10 of 0.969 and Hit Rate@3 of 0.960 through a Retrieve-Reason-Rank-Reflect (4R) pipeline. The system handles zero-history (cold-start) and cross-domain scenarios without persistent state. The framework is fully containerized with Docker Compose and includes Bruno API collections for one-click reproducibility.

1. Introduction & Research Question

The Contextual Intelligence Gap

Global platforms feel alien to Nigerian users because they ignore local realities: price sensitivity relative to haggling thresholds, hyper-local logistics, and linguistic markers that encode trust. Existing recommendation systems rely on persistent Western interaction histories and struggle with zero-history scenarios in high-volatility markets.

Research Question

How can we model inflation-sensitive, culturally-linguistic decision-making for recommendations without persistent user state, while handling zero-history and cross-domain scenarios?

Our answer is stateless archetype reasoning: instead of learning from transaction logs, we simulate the user's internal monologue through Chain-of-Thought (CoT) reasoning. We predict not just what they buy, but why—by modeling economic anxiety and situational urgency as explicit parameters. This eliminates cold-start latency and guarantees reproducibility.

2. Related Work

Prior efforts in agentic AI (like P5 and ReAct) focused on unifying reasoning and retrieval, but assumed persistent profiles and Western corpora. No prior work addresses the specific challenges of cold-start users in high-volatility emerging markets or cross-domain bridging for localized content. We close this gap by proving demographic archetypes plus situational context achieve NDCG@10 = 0.969 without any database, enabling zero cold-start penalty.

3. Task B: Recommendation Agent

3.1 The 4R Pipeline

Retrieve: FAISS (IndexFlatIP, all-MiniLM-L6-v2) filters a top-100 candidate pool, enriched with archetype triggers (e.g., "luxury" for "Big Woman").

Reason: CoT identifies the intersection of occasion (e.g., "Wedding") and location (e.g., "Abuja").

Rank: Symbolic ranker applies Location Boost (+15.0 for city-match), Persona-Interest Alignment (matching interests prioritized), and Cold-Start Inference (demographic best-guesses).

Reflect: Generates personalized "Why" justifications for the top 3 recommendations.

3.2 Cold-Start & Cross-Domain Reasoning

Cold-Start Protocol

For zero-history users, we infer preferences from layered signals:

Archetype Signals (Primary): Haggler biases toward value/budget; Big Woman toward luxury; Community Validator toward popular/trending.

Onboarding Heuristics (Secondary): Lagos users show higher street-food price sensitivity; budget constraints lock value-tier categories.

Implementation: ColdStart.adjust() applies per-item boosts: price-ratio penalties, interest-match bonuses (+5.0), and status-conscious premiums (+3.0).

Cross-Domain Mechanism

All items store cross_domain_tags as semantic bridges. For example, "The Wedding Party" (Nollywood) and "Jollof Rice Catering" (street food) both carry social and celebration tags. The ranker detects overlap and applies a +0.8 boost for occasion-match scenarios.

Multi-Turn Conversational Handling

Conversation history is processed as input context, not stored state. Explicit rejections trigger penalties: "Too far" → −10.0 location penalty; "Too expensive" → −8.0; "Already tried" → −15.0 soft-block. Statelessness is maintained by re-computing all adjustments per request.

4. Data Engineering

4.1 Sources & Scale

AnD Nigerian Seed: 126 hand-crafted products localized to Lagos, Abuja, and Port Harcourt.

Kaggle Yelp Academic Dataset: Ingested for semantic diversity.

The 173-item catalog is our high-fidelity Nigerian validation set. Ablation degradation (NDCG drops to 0.955 when cold-start is removed) proves the model generalizes rather than memorizing.

4.2 Cleaning Pipeline

Archetype Inference: Keyword heuristics map raw Yelp reviews to AnD archetypes.

Unified Embedding Space: FAISS index rebuilt over the merged corpus for seamless cross-domain retrieval.

5. Engineering, Observability & Reproducibility

5.1 System Architecture

The framework is 100% stateless: no databases, no sessions. Deterministic seeding ensures reproducible outputs. Real-time reasoning chains stream via Server-Sent Events (SSE). The hybrid architecture combines FAISS speed with LLM nuance.

5.2 Deployment & Testing

The stack is containerized via Docker Compose:

REST API (Task B): localhost:8001

Bruno collections: and-bruno-doc/ for one-click API testing.

Frontend: and-frontend (React) provides an interface for recommendation testing.

Quick start:

git clone https://github.com/jamesadewara/AnD-workspace.git && cd AnD-workspace/AnD-task-b

docker compose up --build

6. Experimental Results

6.1 Performance Scorecard (Task B)

Metric	Value	Target	Status

NDCG @ 10	0.969	> 0.85	PASS
Hit Rate @ 3	0.960	> 0.80	PASS
Diversity (Cat)	3.0	> 2.0	PASS
Cold-Start Hit Rate	0.872	> 0.70	PASS

6.2 Ablation Study (Task B)

Component Removed	Task B (NDCG)	Impact
Full System	0.969	Baseline
Location Boost	0.978	Diversity drops 30%
Cold-Start Logic	0.955	New-user Hit Rate collapses
Cross-Domain Tags	0.961	Cross-domain recs fall 45%

7. Limitations

The evaluation set (173 items) is focused; scaling to 5,000+ items would strengthen generalization claims. The Reflect step adds ~3–5s absolute latency—mitigated by SSE streaming but requiring caching at production scale.

8. Conclusion

AnD AI (Task B) demonstrates that cultural intelligence is a measurable technical advantage in recommendation. By modeling the Nigerian consumer as an economically-rational, linguistically-situated agent, we achieve state-of-the-art performance without persistent user data. The system is production-ready, fully containerized, and documented for immediate evaluation via AnD-task-b and the companion and-bruno-doc.